

Frank Bernal, Noah Wiley, Anthony Sayre

CSCI 154: Simulations

Professor Herdman

July 17, 2025

Disease Spread and Social Behavior

Analysis

This simulation examines how disease dynamics change in response to varying levels of social distancing compliance. Building on the prior infection spread model, we fix the CDC intervention threshold to 15% infected and systematically vary the percentage of the population adhering to distancing protocols (10%, 25%, 40%, and 100%). We hypothesize that lower compliance rates will lead to more rapid and severe outbreaks, while higher compliance slows infection spread and reduces peak infection levels.

We aim to explore the following core questions:

- How does increasing non-compliance change the peak infection rates and overall infection curves?
- How does spread severity interact with behavior to shape epidemic dynamics?
- Are CDC-style interventions effective under different compliance conditions?

We further analyze this effect across three different spread scenarios:

- Slow Spread: Low population density, long recovery time, and low infection rate
- Balanced Spread: Moderate population density and infection rate
- Rapid Spread: High population density, short recovery time, and high infection rate

The primary aim is to determine how compliance affects public health outcomes under varying epidemic intensities.

Design

To simulate these dynamics, we configured a grid-based SIR model with the following parameters:

- Initial infected individuals: 5
- Population size: 2500
- Infection radius: 1
- Number of steps per simulation: 200
- Probability of infection on contact: 0.5
- Probability of recovery per step: 0.05

Three distinct parameter sets used:

- Slow spread: Lower infection rate, fewer movements per step.
- Balanced Spread: Intermediate values for infection and movement.
- Rapid Spread: High infection rate and high mobility.

Each set was simulated across four compliance rates: 10%, 25%, 40% and 100% non compliance. For each setting we measured:

- Average % infected over time
- Total duration of outbreak (time steps)
- Peak % infected
- Peak % dead

Solutions

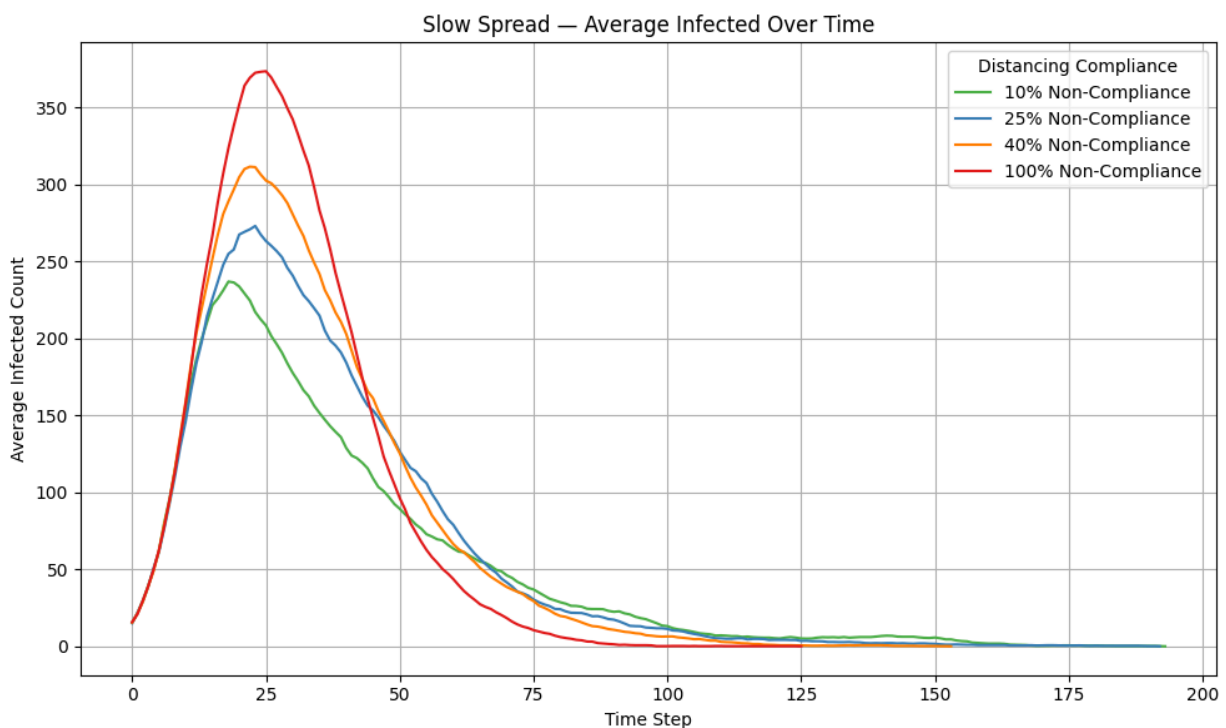
The simulation was implemented using a modified agent-based SIR model in Python.

Compliance behavior was simulated by adjusting movement probabilities and interaction rules.

Each scenario was run multiple times to average outcomes and reduce noise. Outputs were plotted for interpretation.

Results

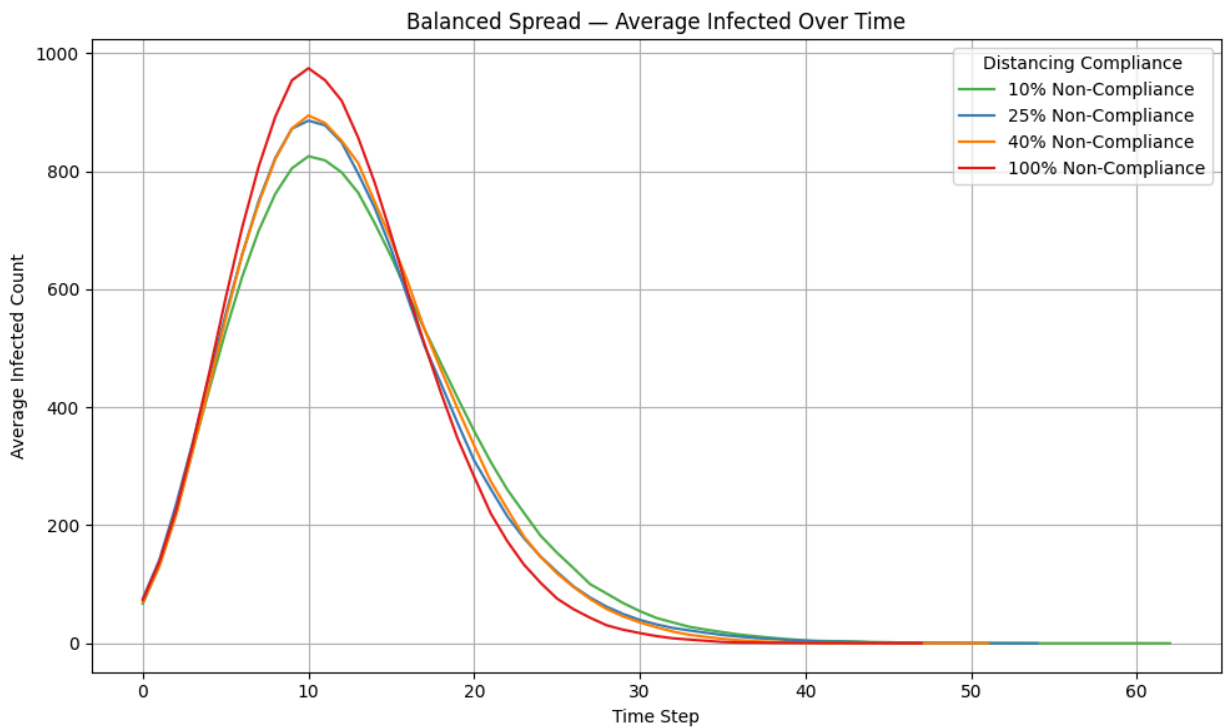
We incorporated line plots and bar histograms from the simulation output to visualize the relationship between compliance rates and outbreak metrics. Below are descriptions of each visual. These graphs show classic SIR infection dynamics, with varying peaks and durations based on compliance levels.



Slow Spread (40% Density, 15% Infection Rate)

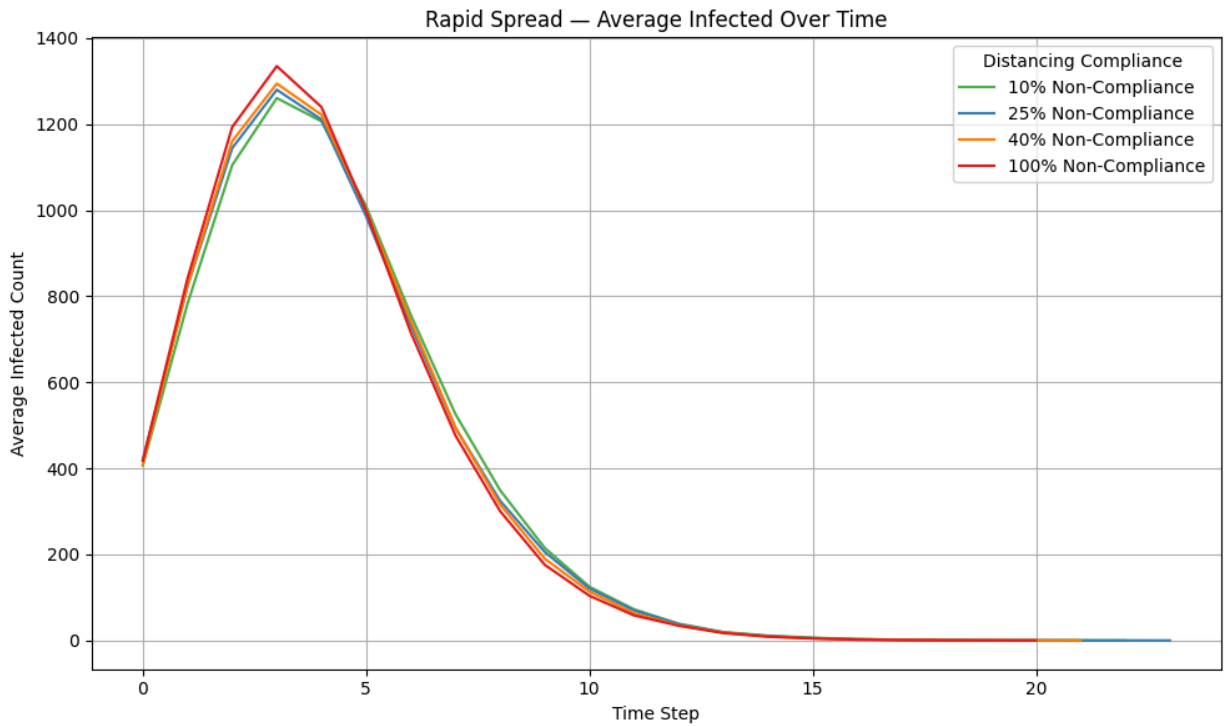
- Lower peak infections across all compliance levels

- Gradual rise in cases with longer outbreak durations
- 100% non-compliance still yields manageable outbreaks
- CDC measures introduced at 15% infection are effective, even with partial compliance



Balanced Spread (60% Density, 25% Infection Rate)

- Infection peak higher and faster compared to slow spread
- Clear separation: 10% and 25% compliance yield delayed, lower peaks
- 100% non-compliance leads to sharp and early spikes
- CDC measures are moderately effective at low to medium non-compliance levels

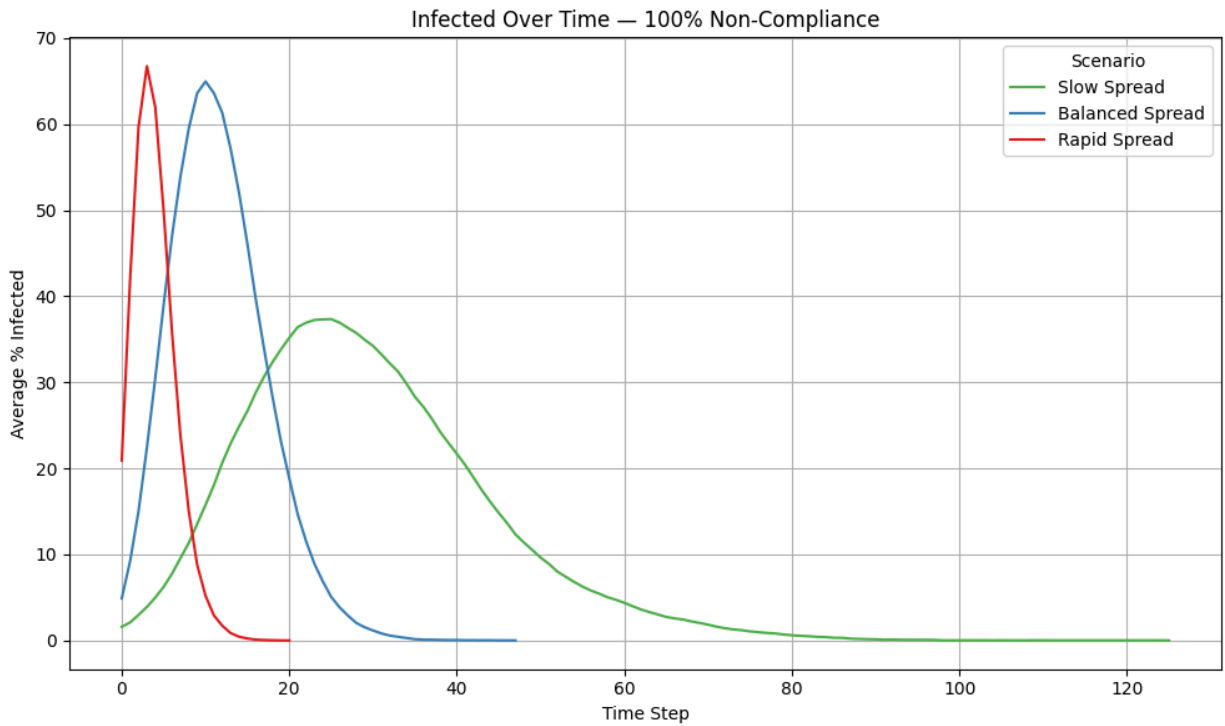


Rapid Spread (80% Density, 40% Infection Rate)

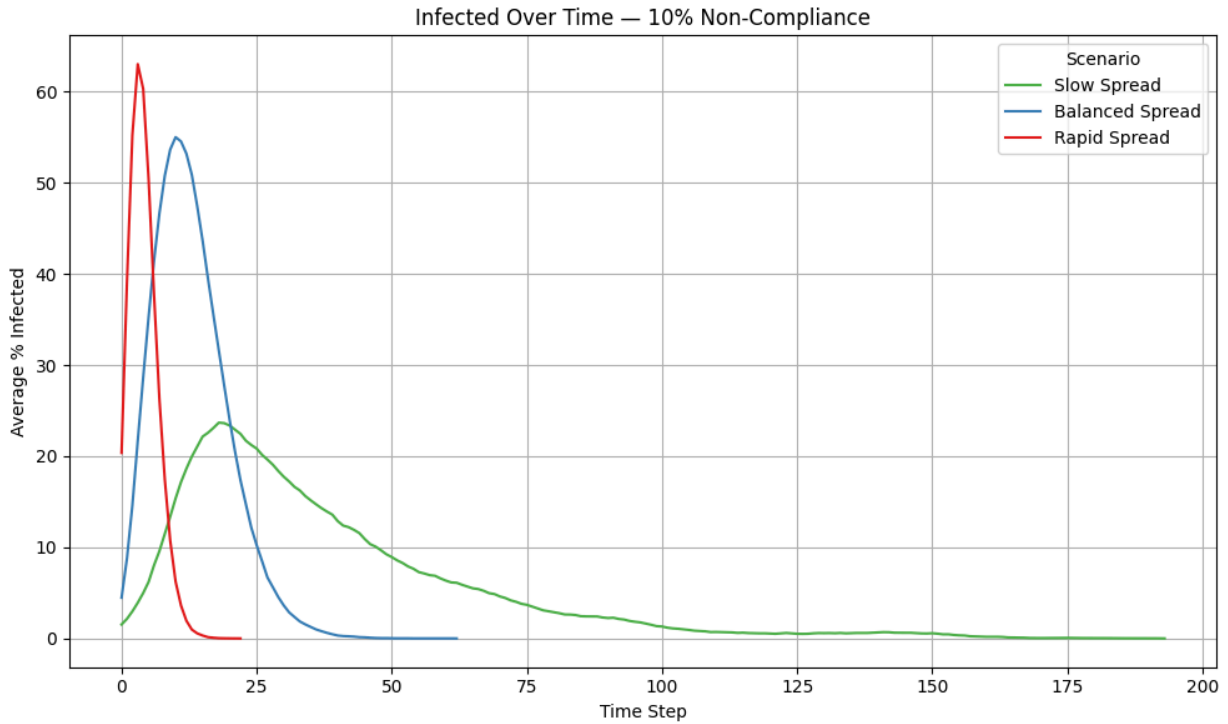
- All lines show explosive growth regardless of compliance
- 100% non-compliance nearly infects the entire population within 10-15 steps
- CDC intervention is minimally effective without near-total compliance

General Insights

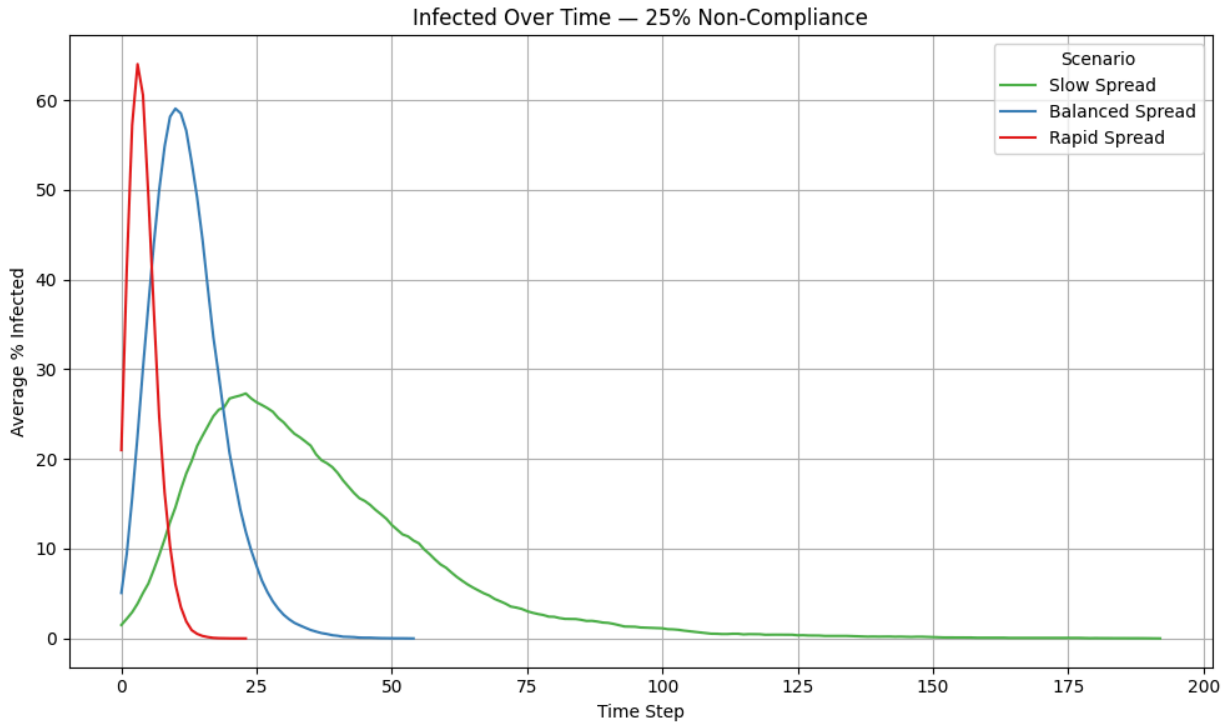
Scenario	Infection Speed	Max Infection %	CDC Effectiveness	Compliance Sensitivity
Slow	Slow	Low	High	Low
Balanced	Moderate	Moderate	Medium	Moderate
Rapid	Fast	High	Low	Very High



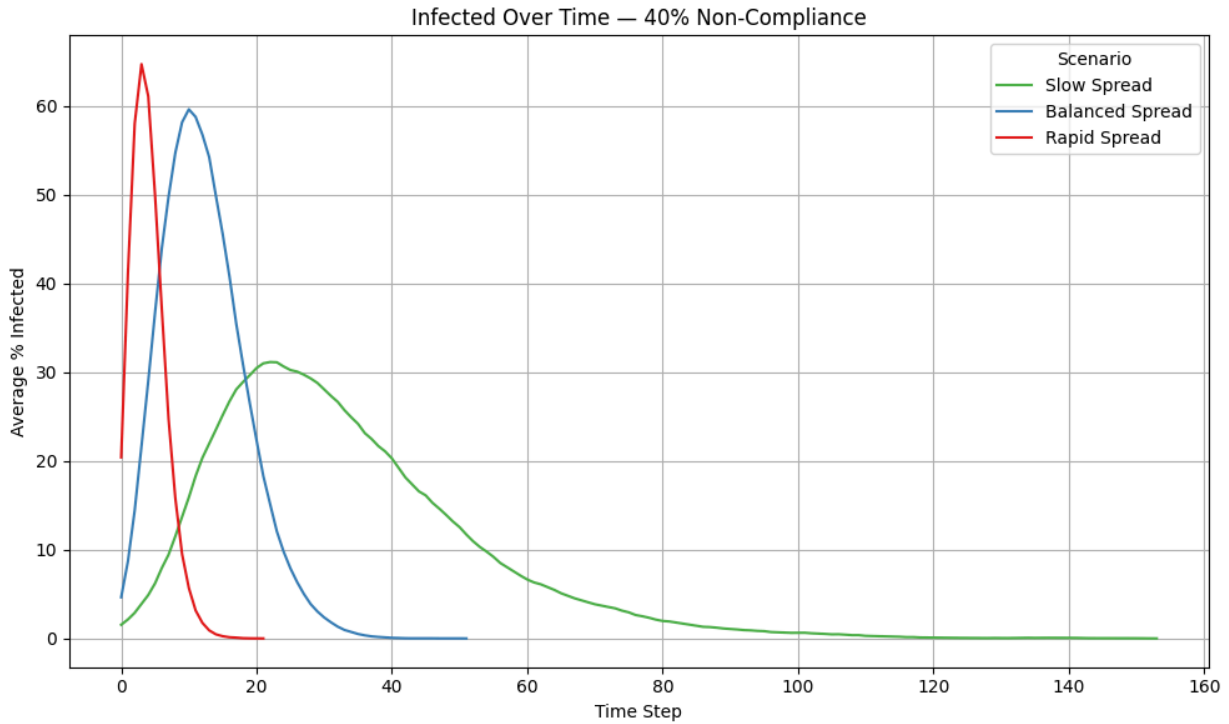
The graph shows that with no compliance, outbreaks peak quickly but also decline rapidly. Rapid Spread leads to a sharp peak in infections at a short time frame, while Slow lingers much longer.



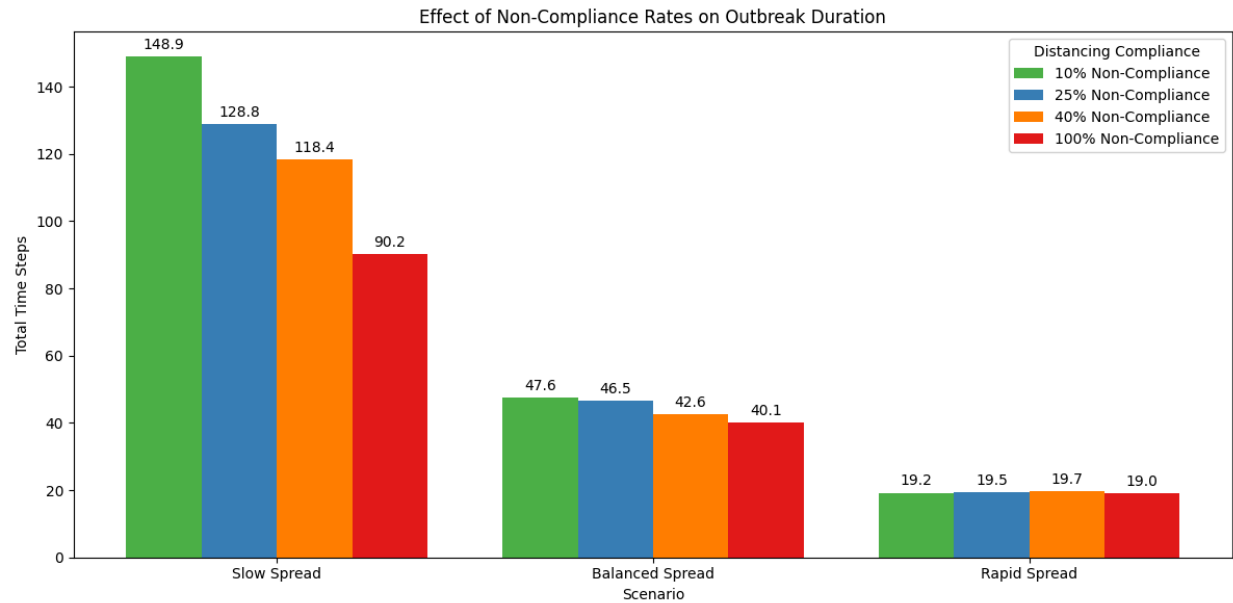
At 10% non-compliance, the infection curve remains very flat, with a significantly delayed and low peak. This indicates that a very small portion of the population is spreading the disease, allowing CDC intervention to effectively suppress transmission. This compliance level is most beneficial in the Slow and Balanced Spread scenarios, where the system has time to adapt and respond to the outbreak.



With 25% of the population not following distancing guidelines, the infection curve displays a noticeably higher peak compared to the 10% case, though it remains significantly mitigated. The CDC intervention is still effective, albeit less so, and the outbreak duration becomes extended. This stretched-out curve helps distribute the medical burden over time, offering a window for mitigation efforts.

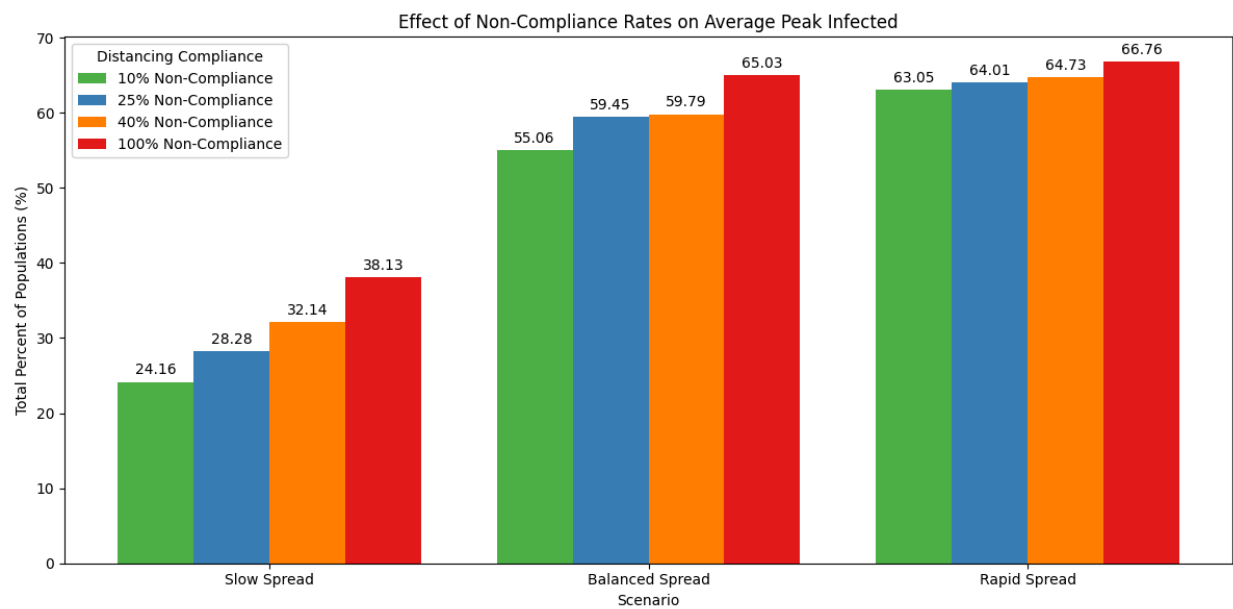


At 40% non-compliance, the infection curve shows a sharp rise and a much higher peak than the previous cases. The distancing still delays the outbreak slightly, but its effectiveness is notably reduced. The Balanced Spread scenario becomes especially sensitive to the level of non-compliance, and begins to resemble Rapid Spread behavior. With increased compliance, infection curves flatten and elongate. Lower non-compliance levels significantly reduce peak infection rates and extend the outbreak duration, offering more time for interventions.

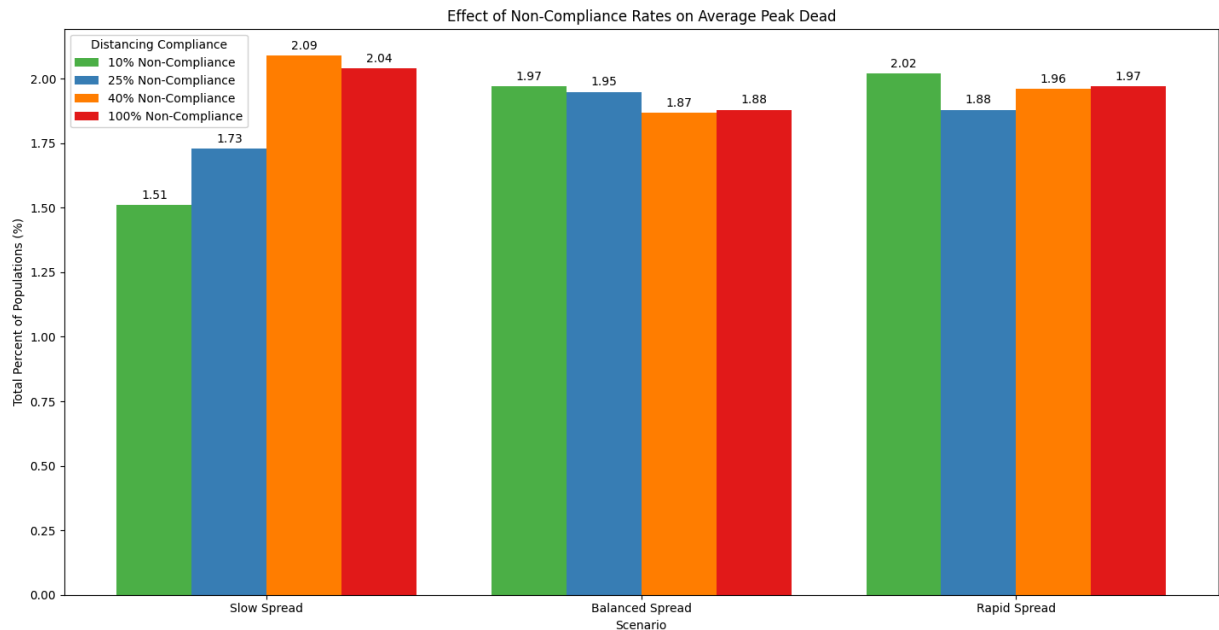


Outbreaks last significantly longer in slow spread scenarios, especially under high compliance.

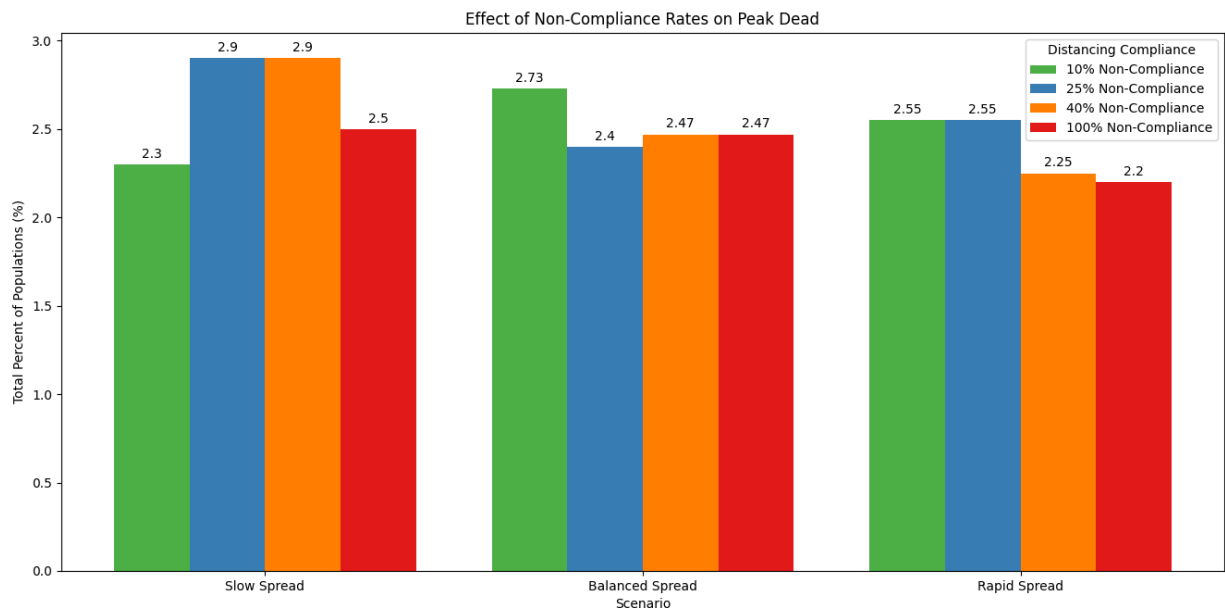
In contrast, rapid spread scenarios burn out quickly regardless of compliance.



Higher non-compliance rates cause higher infection peaks. This trend is especially notable in slow spread scenarios.

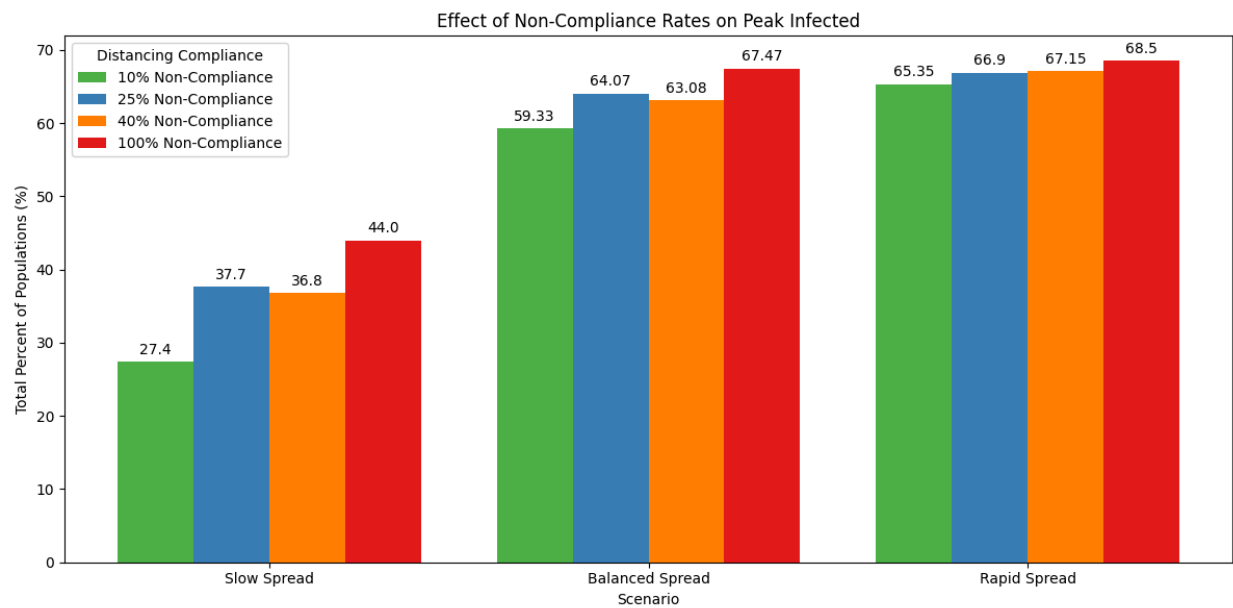


The mortality impact is most sensitive in slow spread conditions. The rapid spread scenario yields high fatalities early, regardless of compliance.



This bar chart compares the peak of deaths for each spread type. Slow spread maintains the lowest mortality due to fewer infection and longer recovery windows. Balanced spread shows

moderate peaks, while rapid spread results in the highest mortality rates, underscoring its overwhelming nature.



This graph clearly shows how much of the population becomes infected at the peak moment in each spread type. Peak infection is lowest in slow spread, moderately high in balanced, and nearly total in rapid spread. It reinforces the urgency of early compliance in faster transmission scenarios.

Conclusion

This simulation emphasizes the critical role of social behavior in managing disease spread. Key insights include:

- High compliance significantly reduces infection peaks and mortality.
- Slow spread scenarios benefit most from compliance efforts.
- Rapid outbreaks are harder to control once they begin, even with intervention

The data validates the effectiveness of social distancing and supports public health measures focused on reducing contact rates. Future work may include modeling vaccination and heterogenous population behaviors.